Ravulapalli Manas Venkata Sai

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Ashoka University, Sonipat, Haryana, India

Research Interests

I work in **computational neuroscience** and **NeuroAI**. I ask how brains and brain-inspired networks learn internal models of the world from their own experience, and how those models support memory, navigation, inference, and planning. My current focus areas are: **cognitive maps and world models**, and how predictive learning from egocentric observation builds them; **factorised representations** in the Tolman–Eichenbaum Machine and successor-representation lineage, which separate the structure of a task from its sensory content so that knowledge transfers to new environments; **active inference and predictive processing**, which treat perception and action as inference under a generative model; **offline replay and sleep**, and how they consolidate and reorganise learned structure; the contribution of **initial random connectivity** and **attractor dynamics** to the representations that emerge; and how environment structure constrains the **representational geometry** of neural populations and models. I do theory-first modelling in close contact with experimental data.

Education

Ashoka University Sonipat, HR, India
B.Sc. (Hons.) Computer Science; minors in Biology and Psychology. GPA 3.66/4.00 Aug 2023 – May 2027 (expected)

- **Thesis:** *Predictive learning folds spatial maps along symmetries.* Supervisor: Prof. Debayan Gupta.

Delhi Public School, Hyderabad, TG (Mathematics, Physics, Chemistry, Computer Science) 2022

Papers and Manuscripts

- **Ravulapalli, M.** (2026). *Predictive learning folds spatial maps along symmetries.* Extended abstract submitted to NeurReps 2026 (NeurIPS workshop); manuscript in preparation for bioRxiv.
  A network that learns a map by predicting its next observation merges places that look the same and that its own movements cannot tell apart; this predicts when place fields repeat across identical rooms. Tested with recurrent networks across environments of varied shape and landmark symmetry.
- **Ravulapalli, M.**, Chadha, S. S., and Hari, A. M. (2026). *Legible Failures: Detecting and Repairing In-Context Binding Errors.* arXiv:2609.11216.
  On wrong answers, a linear probe still recovers the correct binding from the hidden state across 16 models; probe–model disagreement detects failures better than model confidence, and probe-guided steering repairs errors with no label.
- **Ravulapalli, M.**, and Chadha, S. S. (2026). *In-Context Binding Capacity in Language Models.* arXiv:2609.30634.
- **Ravulapalli, M.**, and Chadha, S. S. (2026). *Decodable In-Context State and Model Output Across Training.* arXiv:2609.31401.
- Miglani, A., Chadha, S. S., Li, K., and **Ravulapalli, M.** (2026). *A Heckler in the Hidden State: Correctness Signals in Diffusion Language Models.* arXiv:2609.36783.
- **Ravulapalli, M.**, *et al. Supply-dependent foreclosure: load-dependent loss of plasticity in small language models.* In preparation.

Research Experience

Perseus (Y Combinator-backed) Remote
Chief Scientist *Mar 2026 – Present*

- **Memory representation and retrieval.** Lead research separating what a model internally represents from what it can retrieve and use at decision time, using causal interventions, developmental checkpoints, and attention-routing analyses.
- **Developmental learning and plasticity.** Designed a large controlled training program to measure when computational structures are acquired, maintained, forgotten, and recoverable under changing supervision.
- **Cognitive-map architectures and world models.** Developed MapCore, a persistent-state module with action-conditioned updates, gated memory writes, and imagined multi-step rollouts, so that agents can predict future states and plan over long horizons.
- **Interpretability and representation analysis.** Built tools for probing internal state and information flow, including Jacobian-based binding probes, path patching, checkpoint sweeps, and representation-use dissociations.

Independent Research, Ashoka University Sonipat, HR
Predictive RNNs and cognitive map formation *Aug 2025 – Present*

- Implemented the Levenstein et al. (2024) sequential predictive RNN and tested how topology, geometry, scale, symmetry, and perceptual aliasing shape cognitive maps across ten controlled environments in RatInABox and MiniGrid.
- Built diagnostics for representational structure: occupancy-normalised mutual information, geodesic sRSA, participation ratio, RDMs, and persistent homology.

European Molecular Biology Laboratory (EMBL) Heidelberg, Germany
Summer School Student *Jul 2025*

- Used optical tweezers and kymograph analysis to study how transcription factors locate target DNA sequences; applied bio-image analysis and 3D reconstruction to light-sheet and confocal volumes.

Neural Systems Lab, University of Hyderabad Hyderabad, TG
Summer Research Assistant *Jun – Aug 2025*

- Ran behavioural assays in weaver ants (mirror-recognition replications), insect electrophysiology with rig setup and calibration, and bulk drug tracing in grasshoppers.

Neuro-ethology Lab, Ashoka University Sonipat, HR
Research Assistant *Aug 2024 – Present*

- Developing a behavioural paradigm for numerical cognition in zebrafish. Built automated tracking (OpenCV, YOLOv8) and closed-loop infrastructure for zebrafish virtual reality.

Training and Conferences

CAMP 2026, Computational Approaches to Memory and Plasticity IISER Pune, India
Selected participant *2026*

- Modelled persistent activity in the goldfish oculomotor integrator using NEURON and dynamical-systems analysis; also implemented a Brian2 spiking Q-learning bandit and a Tsodyks–Markram dynamic-synapse network.

Conferences attended: Conference on Neural Information Processing Systems (NeurIPS); Annual Meeting of the Indian Academy of Neurosciences (IAN).

Teaching

Ashoka University Sonipat, HR
Teaching Fellow, Computational and Theoretical Neuroscience *Jan – May 2025*

- Co-designed the course with Prof. Bittu K. Rajaraman, delivered 15 lectures, and ran the journal club. Wrote MATLAB assignments from neural encoding and compartmental models to recurrent and Hopfield networks.

Ashoka University Sonipat, HR
Teaching Assistant, Introductory Electrophysiology *Aug – Dec 2025*

- Designed a hands-on course for 10 students on setup, maintenance, and error checking of an Axon rig for patch clamp and microelectrode recording, with ERG/EAG practicals and Python spike train analysis tutorials.

Selected Projects

- **Cognitive assessment platform for neurodegenerative disorders**: Built a digital battery for attention, working memory, motor deficits, and visuospatial processing in Alzheimer's patients and controls, using convex-hull estimation and constrained path tracing with automated scoring and longitudinal logging.

Technical Skills

Computational neuroscience and NeuroAI: Cognitive maps, recurrent and predictive networks, continuous attractor networks, spiking and biophysical models, reinforcement learning, dynamical systems, transformer interpretability (probing, path patching, steering)
Representation and data analysis: Representational similarity analysis, participation ratio, RDMs, persistent homology, regression, mixed-effects models, Bayesian inference, survival analysis, bootstrap and cluster-aware inference, power analysis
Programming and scientific computing: Python, C++, C, Rust, MATLAB, JavaScript; PyTorch, NumPy, SciPy, statsmodels, PyMC, Brian2, NEURON, NEST, RatInABox, MiniGrid, Ripser, Git, Docker
Neural and behavioural data analysis: SpikeInterface, Kilosort, Open Ephys, Suite2p, SIMA, DeepLabCut, SLEAP, BORIS, Ilastik, Napari; SPM, fMRIPrep, nilearn
Experimental: Behavioural paradigm design, patch clamp, intra- and extracellular microelectrode recording, LFP, ERG/EAG, rig setup and maintenance, dissection, bulk drug tracing